

Lab 3: Robot Motion Control

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Exercise 1

1.1: Kobuki Pathing

Exercise 1 was initiating connection with the Kobuki and having it follow a certain path. This path consisted of a 90 degree turn at the beginning, moving straight forward 50mm and then following a 90 degree bend with a radius of 50mm.

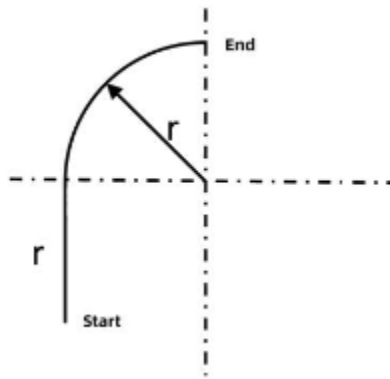


Figure 1: Kobuki Path

Exercise 2

2.1 Joystick Driving

Exercise 2 was about reading the input from the Logitech controller (D-pad) and using that to send data packets with movement information to the Kobuki to manually drive around.



Figure 2: Logitech Controller

Exercise 3

3.1 TCP/IP Server Connection

Exercise 3 was about creating a TCP/IP connection between 2 raspberry Pi's to control the Kobuki.



Figure 3: Raspberry Pi Zero 2W (Wifi Server)

Bonus

1. Full Stack Development

The first bonus was to create both the Client and Server side to control the Kobuki. We did this by using the Raspberry Pi Zero 2W as the server to drive the Kobuki with the commands received from the Raspberry Pi 4B

2. Joystick Control

The second bonus was to implement the joystick instead of the D-Pad to drive the Kobuki robot. We altered our code from the Bonus 1 to read the data from the joystick and send that as a drive command to the Server.

Supplemental Questions

Briefly summarize what you learned from this lab.

In this lab, we learned how to control the Kobuki robot from a program, actively controlling it with a controller, and by passing messages from client to server. We learned how to connect to the Kobuki through our code and fine-tune its movement to function as intended. We also mapped controller inputs to the Kobuki movement and were able to operate it in real time.

Explain the way the Kobuki's movement is controlled.

The Kobuki's movement is controlled by adjusting the speed and radius sent via data packets over USB. With the D-pad, we could make it go forward and backward with up and down and left and right with left and right. When we changed controls to using the joysticks, the left proportionally controlled forward and backward movement, and the right proportionally controlled increases in left or right turn radius.

Explain the steps of a complete control request from the client to the server.

The client side polls the Logitech controller inputs using `joystick.sample`, it then packages this data into a comma separated string, this string is then sent via sock to the server, when the server receives the message, it reads it into the buffer. After the message is done sending and in the buffer, we then parse the values into variables for Vertical, Horizontal, stopping, and closing the connection events. The server then creates a data packet for the Kobuki and sends that over USB to make the Kobuki move.

ACKNOWLEDGMENTS

I certify that this report is my/our own work, based on my/our personal study and/or research and that I/we have acknowledged all material and sources used in its preparation, whether they be books, articles, reports, lecture notes, and any other kind of document, electronic or personal communication. I/We also certify that this assignment/report has not previously been submitted for assessment anywhere, except where specific permission has been granted from the coordinators involved.

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REFERENCES

- 1) Lab 3 Slides
- 2) Lab 3 Handout
- 3) Lab 3 Documents